

DESIGN AND DEVELOPMENT OF AN INTEGRATED PASSIVE COOLING SYSTEM FOR RESIDENTIAL BUILDINGS

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

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ABSTRACT

The rapid growth of urbanization and the extensive use of air conditioners have led to a serious issue with respect to indoor thermal comfort in a residential building with hot and tropical climates. The traditional air-conditioning systems require massive amounts of electrical energy which also contribute to the rise in greenhouse gases. Hence, it becomes essential to find cost-effective and sustainable methods to reduce indoor temperatures, improving thermal comfort, and reducing greenhouse gas emissions.

This project aims at designing and analyzing passive cooling systems to reduce indoor temperatures and enhance thermal comfort without the conventional usage of air-conditioning systems. The project studies various passive cooling techniques like cross and stack ventilation, cool roof technology, use of shading devices, consideration of building orientation, and the inclusion of thermal mass.

The analysis of these passive cooling techniques is carried out by considering fundamental modes of heat transfer such as conduction, convection, and radiation. The effect of external factors like ambient temperature, solar radiation, wind velocity, and relative humidity on the performance of these passive techniques has also been considered.

The project reports that while the individual use of passive techniques can reduce indoor temperatures by 1-4 °C , the combined use of these techniques can reduce the indoor temperatures by 4-6 °C . The project emphasizes the significance of passive cooling systems by highlighting the role of appropriate design and climatic considerations to achieve enhanced thermal comfort while reducing energy consumption, thus being a cost-effective and sustainable option.

TABLE OF CONTENT

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	iii
	LIST OF TABLES	vi
	LIST OF FIGURES	vii
1	INTRODUCTION	1
	1.1 Methodology	2
2	LITERATURE REVIEW	4
3	COMPONENTS, SPECIFICATION AND DESCRIPTION	7
	3.1 Components List	7
	3.2 Component Description	7
	3.2.1 RTC Module	7
	3.2.2 SD Card Module	8
	3.2.3 ESP32 Development Board	8
	3.2.4 OLED Display	9
	3.2.5 Memory Card (8 GB)	9
	3.2.6 CR2032 Coin Cell	10
	3.2.7 DS18B20 Temperature Sensor	10
	3.2.8 Jumper Wires	10
	3.2.9 USB Data Cable	11
	3.2.10 Cardboard	11
	3.2.11 Glue Gun	12
	3.2.12 Aluminium Foil	12
	3.3 CAD Model Diagram	13

CHAPTER NO.	TITLE	PAGE NO.
	3.4 Block Diagram	14
4	WORKING PRINCIPLE	15
	4.1 Heat Gain Mechanism	15
	4.2 Passive Cooling Techniques Applied	15
	4.3 Natural Ventilation (Cross & Stack Effect)	16
	4.4 Heat Reduction Using Cool Roof and Shading	16
	4.5 Thermal Mass Effect	17
	4.6 Temperature Monitoring System	17
	4.7 Working of ESP32 Based Monitoring System	18
	4.8 Passive Prototype - Calculations	21
5	COST ESTIMATION	24
6	RESULTS AND SYSTEM OUTPUTS	25
	6.1 Temperature Variation Results	25
	6.2 Sensor Readings and Data Logging	26
	6.3 Result Tables and Graphs	27
7	SCOPE OF FUTURE WORK	30
	7.1 Advantages	30
	7.2 Applications	31
8	APPENDICES	32
	8.1 Field Visit	32
	8.2 Photographs of Prototype Setup	34
	8.3 Arduino Code for Temperature Logging	35
9	CONCLUSION	36
	REFERENCES	37

LIST OF TABLES

TABLE NO.	TITLE	PAGE NO.
Table 3.1	List of Components and Specifications	7
Table 4.1	Passive Cooling Techniques Comparison	19
Table 4.2	Heat Transfer Parameters	20
Table 5.1	Cost Estimation	24
Table 6.1	Temperature Variation Results	25
Table 6.2	Sensor Readings and Data Logging	26
Table 6.3	Temperature Difference (ΔT) Analysis	28

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE NO.
Figure 1.1	Working Methodology (Flow Chart)	3
Figure 3.1	RTC Module	8
Figure 3.2	SD Card Module	8
Figure 3.3	ESP32 Module	9
Figure 3.4	OLED Display	9
Figure 3.5	Memory Card (8GB)	9
Figure 3.6	CR2032 Coin Cell	10
Figure 3.7	DS18B20 Temperature Sensor	10
Figure 3.8	Jumper Wire	11
Figure 3.9	USB Data Cable	11
Figure 3.10	Cardboard	12
Figure 3.11	Glue Gun	12
Figure 3.12	Aluminium Foil Sheet	12
Figure 3.13	CAD Diagram	13
Figure 3.14	System Block Diagram	14
Figure 6.1	Temperature Variation Graph	29
Figure 8.1	Field Visit Images	35
Figure 8.2	Prototype Setup House	35

CHAPTER 1

INTRODUCTION

Rapid urbanization and growing need for energy for space cooling have made thermal comfort in residential buildings a key issue. The large use of conventional air conditioners for space cooling leads to consumption of huge amounts of electricity and emanation of greenhouse gases. This has led to a need for more sustainable and energy efficient methods for maintaining thermal comfort in buildings.

Passive cooling methods are effective in reducing the temperature of residential buildings by reducing heat gains and enhancing heat losses out of the buildings. They are a good alternative to the use of air -conditioning systems in residential buildings.

This project involves designing a passive cooling system and analyzing its effectiveness in reducing temperatures inside buildings. A prototype model is set up using easily available and affordable materials. Temperature sensors are used to record temperatures inside the building at different times. The data logging system is designed using a real time clock and a SD card module. The results are analyzed to examine the effectiveness of each passive method of cooling.

The project aims at analyzing the effectiveness of passive cooling models in reducing temperatures inside residential buildings. It demonstrates the applicability of passive cooling systems in residential buildings as a more sustainable and energy efficient alternative to conventional air -conditioning systems.

The proposed passive cooling prototype is equipped with a temperature monitoring system to evaluate its thermal performance. Two DS18B20 digital temperature sensors are used to measure the indoor and outdoor temperatures. An ESP32 development board acts as the main controller for acquiring the sensor data. An OLED display is provided for real-time visualization of temperature values, while an RTC module provides time information and an SD card module is used for data logging. This integrated monitoring system enables the temperature performance of the passive cooling prototype to be observed and evaluated without depending continuously on a laptop.

1.1 METHODOLOGY

The methodology adopted for this project is comprised of several steps for designing, developing and evaluating a passive cooling system for residential buildings. The methodology considers the integration of appropriate passive cooling techniques and monitoring of temperature variations to analyse the performance of the developed prototype.

A literature survey was initially conducted to explore passive cooling concepts and techniques such as natural ventilation, cool roof, shading and thermal mass. The survey enabled the identification of appropriate passive building cooling techniques that were integrated during the development of the model. A prototype of a residential building was designed and developed using cardboard as a primary material.

Aluminium foil was attached to specific surfaces and roof sections in order to reduce heat gain through radiations. Selected surface openings were made in order to allow cross and stack venting of the model. The developed prototype reflected the basic features of a residential building subjected to outdoor climatic conditions. Temperature sensors (DS18B20) were used to monitor and record temperature variations at specific model locations. The RTC module was included for recording time and date information while an SD card module that was attached to an 8 GB memory card was used to store the collected data. The components were connected using a jumper wire and the circuit was connected to a computer using a USB data cable.

The developed prototype was subjected to outdoor conditions in order to estimate the performance of the developed passive cooling system. The indoor and outdoor temperatures were monitored and recorded at specific time intervals. The recorded temperature data was analysed and reported in tables and graphs in order to illustrate temperature variation patterns and evaluate the performance of the developed passive cooling system.

The methodology comprised of a literature study, selection of passive cooling techniques, model design and fabrication, component integration and temperature recording, data collection and analysis.

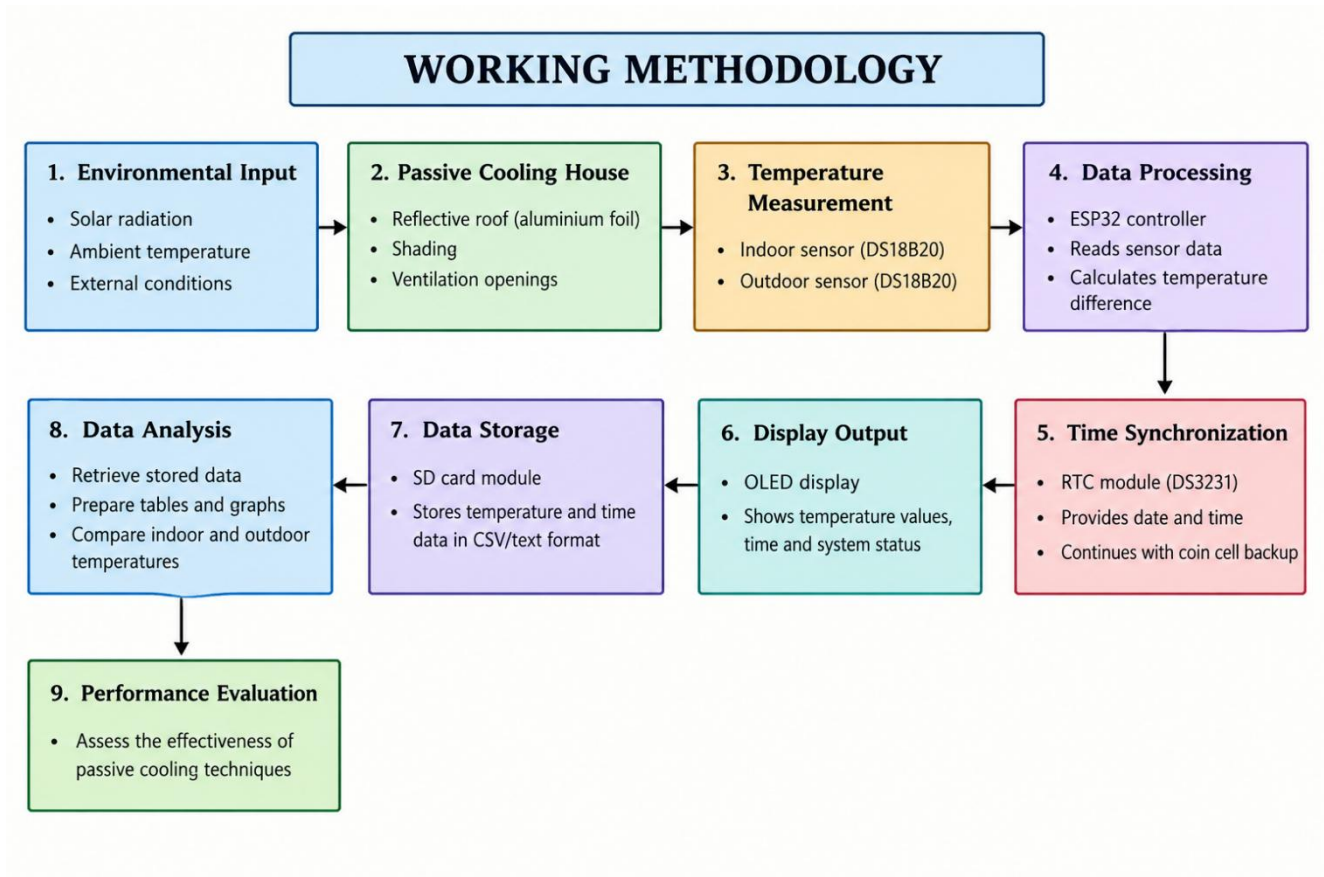


Figure 1.1 Working methodology

CHAPTER 2

LITERATURE SURVEY

Givoni (1998)

Givoni (1998), in *Climate Considerations in Building and Urban Design*, discussed the importance of climate-responsive building design for achieving thermal comfort with reduced dependence on mechanical cooling systems. The work explains how building orientation, natural ventilation, solar control, building materials, and other passive design strategies can be selected according to local climatic conditions. The study emphasizes that proper building design can reduce unwanted heat gain and improve natural heat dissipation. These concepts are relevant to the present project because the proposed passive cooling prototype incorporates natural ventilation, reflective roofing, shading, and suitable building design to reduce indoor temperature.

Olgyay (1963)

Olgyay (1963), in *Design with Climate: Bioclimatic Approach to Architectural Regionalism*, presented the importance of designing buildings according to the surrounding climatic conditions. The work focuses on the relationship between climate, building form, solar radiation, air movement, and human thermal comfort. It highlights the role of orientation, shading, ventilation, and appropriate architectural design in controlling the indoor thermal environment. These principles provide a theoretical foundation for the present project, where ventilation openings, roof design, and shading are incorporated into the residential prototype to minimize heat gain and improve indoor comfort.

Santamouris (1994)

Santamouris (1994), in *Passive Cooling of Buildings*, discussed various passive cooling approaches used to reduce indoor temperatures and cooling energy requirements. The work describes techniques such as natural ventilation, solar control, evaporative cooling, and thermal storage as important methods for improving building thermal performance. Passive cooling techniques utilize natural heat transfer and air

movement instead of continuously operating mechanical cooling equipment. The study is relevant to the present project because the proposed prototype combines natural ventilation, reflective roofing, shading, and thermal mass as integrated passive cooling techniques.

Szokolay (2014)

V. Szokolay (2014), in *Introduction to Architectural Science: The Basis of Sustainable Design*, explained the importance of sustainable and climate-responsive building design. The work considers heat transfer, solar radiation, ventilation, building materials, and thermal performance as important factors in achieving comfortable indoor conditions. Proper selection of building elements and passive strategies can help control heat flow through the building envelope. This provides useful guidance for the present project, particularly in understanding heat gain through the roof and walls and the use of reflective materials and ventilation openings in the prototype house.

Nayak and Prajapati

K. Nayak and J. A. Prajapati, in *Handbook on Energy Conscious Buildings*, discussed energy-efficient approaches for building design and the importance of reducing unnecessary energy consumption. The work emphasizes the use of climate-responsive design strategies and passive techniques to improve building performance. Natural ventilation, solar control, appropriate building materials, and other energy-conscious measures can reduce the requirement for conventional cooling. These principles support the objective of the present project, which aims to demonstrate a low-cost passive cooling approach for residential buildings while reducing dependence on mechanical cooling systems.

Krishan et al.

Krishan, N. Baker, S. Yannas, and other contributors, in *Climate Responsive Architecture*, presented concepts related to climate-responsive building design and passive environmental control. The work explains how architectural features can be

adapted to local climatic conditions to control solar heat gain, air movement, and indoor thermal conditions. Natural ventilation, shading, building orientation, and appropriate construction techniques are important elements of climate-responsive architecture. These concepts are applied in the present project through ventilation openings, reflective roof treatment, shading, and the selection of suitable prototype materials.

Bureau of Energy Efficiency (BEE) (2017)

The Bureau of Energy Efficiency (BEE), through the Energy Conservation Building Code (ECBC), 2017, provided guidelines related to energy-efficient building design and reduction of energy consumption. The guidelines emphasize the importance of building envelope performance, solar heat control, and energy-conscious design practices. Such approaches help reduce the cooling demand of buildings and improve overall energy efficiency. The principles are relevant to the present project because passive cooling techniques are used to reduce heat gain before relying on conventional mechanical cooling systems.

CHAPTER 3

COMPONENTS, SPECIFICATION AND DESCRIPTION

3.1 COMPONENTS LIST

S.No	Component Name
1	RTC Module
2	SD Card Module
3	ESP 32 Development Board
4	OLED Display
5	Memory Card
6	CR2032 Cell
7	Temperature Sensor
8	Jumper Wires
9	USB Data Cable
10	Cardboard
11	Glue gun
12	Aluminium Foil

Table 3.1 List of components

3.2 COMPONENTS DESCRIPTION

3.2.1 RTC Module (Real Time Clock Module)

The RTC module is utilized to store accurate time and date information during the operation of the system. The system timestamp enables the storage of temperature data with time information. In addition, the RTC works continuously even when the main system is turned off by using a coin cell as a power supply. This feature is required for long-term temperature cycle analysis and logging in passive cooling experiments.

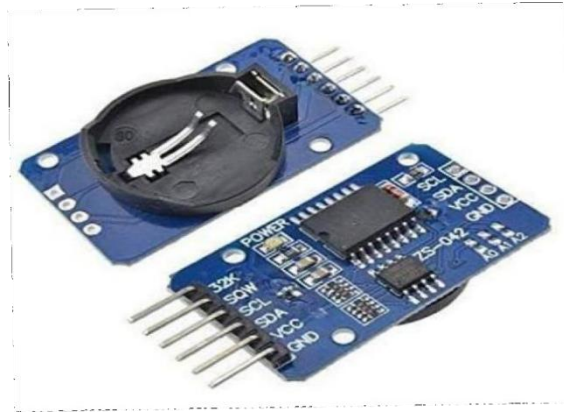


Figure 3.1 RTC Module

3.2.2 SD Card Module

The SD card module provides an interface between the microcontroller and the memory card. It allows storing of the temperature data acquired from the temperature sensors on the card in a human-readable format. The module also enables serial communication to read back the stored data for analysis of the cooling trends over time.



Figure 3.2 SD Card Module

3.2.3 ESP32 Development Board

The ESP32 development board is the main controller of the temperature monitoring system. It receives temperature sensor readings from the DS18B20 sensors and processes the data. In addition, it communicates with the RTC, OLED, and SD card to receive time information, display temperature readings, and store the data for additional analysis.

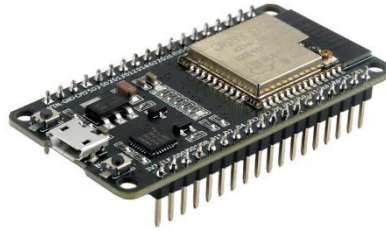


Figure 3.3 ESP 32 Module

3.2.4 OLED Display

The OLED display provides a real-time display of the collected temperature data. It displays the indoor temperature, outdoor temperature, temperature difference, and other relevant information received from the ESP32. This information allows the prototype to display data, which is useful when demonstrating the system without having to keep a laptop connected to the system.



Figure 3.4 OLED Display

3.2.5 Memory Card (8 GB)

The 8 GB memory card is used to store the large amount of temperature and time data collected during the experiment. It is a good size for this particular experiment because the data collected can be put onto a computer and graphed to show passive cooling results.



Figure 3.5 Memory Card (8GB)

3.2.6 CR2032 Coin Cell

The CR2032 coin cell supplies backup power to the RTC module. It ensures that time and date information are retained even when the main power supply is disconnected. This helps maintain accurate data logging without time errors during power interruptions.



Figure 3.6 CR2032 Coin Cell

3.2.7 DS18B20 Temperature Sensor

The DS18B20 is a digital temperature sensor used to measure ambient and indoor temperatures of the passive cooling setup. It provides accurate readings over a wide temperature range and supports one-wire communication, allowing multiple sensors to be connected using a single data line. This sensor is suitable for monitoring temperature variations caused by passive cooling techniques.



Figure 3.7 DS18B20 Temperature Sensor

3.2.8 Jumper Wires

Jumper wires are used to establish electrical connections between modules, sensors, and the microcontroller. They provide flexibility in circuit assembly and are ideal for prototype development without soldering.



Figure 3.8 Jumper Wires

3.2.9 USB Data Cable

The USB data cable is used to power the microcontroller and upload program code from the computer. It also enables serial communication for debugging and real-time monitoring of sensor data during experimentation.



Figure 3.9 USB Data Cable

3.2.10 Cardboard

Cardboard is used to construct the physical model of the building structure. It represents walls, roofs, and partitions of the passive cooling prototype. Cardboard is lightweight, low-cost, and easy to modify, making it suitable for experimental thermal studies.



Figure 3.10 Card Board

3.2.11 Glue Gun

The glue gun is used to assemble and fix structural components of the prototype securely. It provides strong bonding between cardboard parts and helps maintain structural stability during experimentation.



Figure 3.11 Glue Gun

3.2.12 Aluminium Foil Sheet

Aluminium foil is used as a reflective surface to reduce heat gain due to solar radiation. When applied to roof or wall surfaces, it reflects a significant portion of incident solar energy, thereby improving the passive cooling performance of the model.



Figure 3.12 Aluminium Foil Sheet

3.3 CAD MODEL DIAGRAM

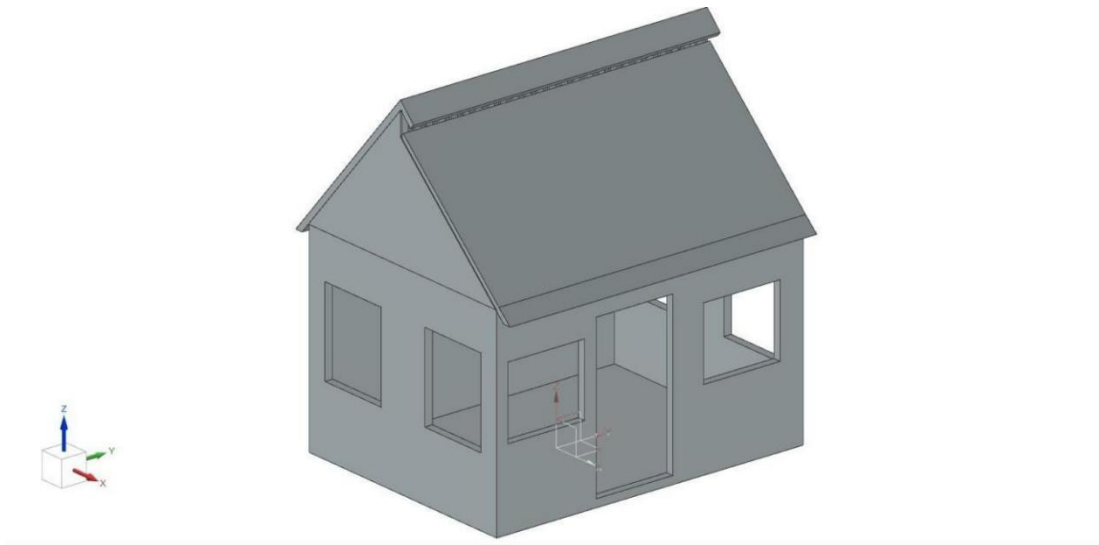


Figure 3.13 Cad Diagram

The CAD model of the passive cooling prototype house was developed to visually represent the structural layout and integration of various passive cooling techniques used in the project. The model illustrates the overall geometry of the house, roof configuration, ventilation openings, shading elements, and placement of temperature sensors.

The prototype consists of lightweight wall construction using cardboard material, a sloped roof arrangement covered with aluminium foil to reflect solar radiation, and strategically placed ventilation openings to enable cross and stack ventilation. The CAD model also highlights the airflow path from inlet openings to outlet vents, which is essential for natural ventilation.

This model helps in understanding heat gain mechanisms, airflow movement, and thermal behaviour of the structure before physical fabrication. It also serves as a reference for construction, analysis, and performance evaluation of the passive cooling system.

3.4 SYSTEM BLOCK DIAGRAM

The system block diagram represents the overall working flow of the proposed passive cooling and temperature monitoring system. It shows the sequential interaction between the environmental conditions, passive cooling techniques, temperature measurement, ESP32 controller, RTC module, OLED display, SD card storage, and data analysis. The diagram provides a clear overview of how temperature data is measured, processed, displayed, stored, and finally used to evaluate the performance of the passive cooling system.

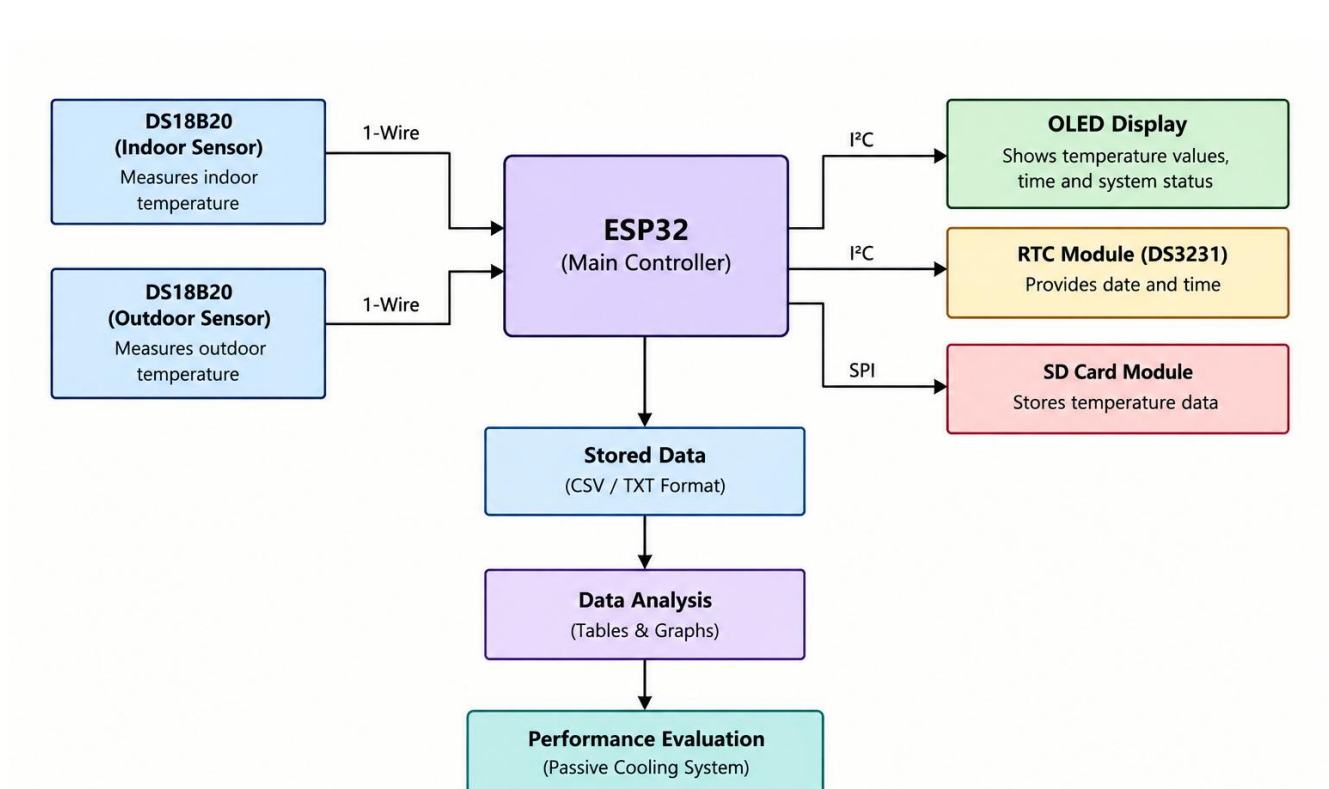


Figure 3.14 System Block Diagram

CHAPTER 4

WORKING PRINCIPLE

4.1 HEAT GAIN MECHANISM

Heat gains in residential buildings mainly arise from solar radiation, conduction through materials and internal heat sources. In warm and hot climates, the roof and walls contribute the most to heat gains in a building since they are constantly exposed to solar radiation. Solar radiation hitting the roof and wall surfaces heats them up and the indoor space is warmed up through conduction.

Moreover, heat is transferred indoors when warm air gains access to the building interior and inadequate cooling of the building interior takes place. Heat is also generated indoors by occupants and electrical appliances.

The cooling process is significant in determining the most appropriate passive design strategies that can be employed in a building to reduce the amount of heat gained and enhance cooling of the building interior.

The effect of heat gain is evaluated experimentally by measuring the indoor and outdoor temperatures of the prototype. The temperature difference between these two locations provides an indication of the thermal performance of the passive cooling arrangement.

4.2 PASSIVE COOLING TECHNIQUES APPLIED

The passive cooling technique employed in the project is a combination of various passive cooling technologies to reduce heat gains in the building structure. The project employs the use of natural ventilation, cool roof, shading and thermal mass. Natural ventilation allows for the circulation of cool air inside the building which lowers the indoor temperature by removing stale hot air inside the building. Cool roof and reflective surfaces prevent solar gains by reflecting solar energy away from the building structure. Shading blocks the penetration of solar radiation into the building interior.

The combined effect of the passive cooling techniques enhances the passive design of the prototype house by reducing solar and thermal gains.

In addition to the passive cooling features, an electronic temperature monitoring system is integrated into the prototype. The monitoring system does not provide active cooling; instead, it is used only to measure and record the thermal response of the passive cooling design.

4.3 NATURAL VENTILATION (CROSS & STACK EFFECT)

Natural ventilation is critical in enhancing passive cooling of the building interior. Natural convection or stack effect is used to cool the building interior by facilitating the circulation of air inside the building. In cross ventilation, air circulation takes place as air enters through one opening and exits through another opening at the opposite side of the building. The stack effect works by exhausting hot air at the top of the building and allowing cool air to circulate at the lower part of the building interior.

Ventilation openings are provided on the prototype to allow air movement through the building. Air entering through one opening can move across the indoor space and exit through another opening. Higher-level openings can also assist the escape of warm air through the stack effect.

4.4 HEAT REDUCTION USING COOL ROOF AND SHADING

The roof is one of the main sources of heat gain in building structures. In this project, cool roof technology has been used to prevent solar gain. The aluminium foil reflective layer applied over the roof reflects solar energy and reduces the amount of heat transferred into the building interior.

Shading has also been used to prevent solar radiation from penetrating through the openings in the building structure. A shading device reduces the amount of solar radiation that would have penetrated the building interior thus reducing solar heat gains. The cool roof and shading techniques reduce the amount of heat gained by the prototype house.

4.5 THERMAL MASS EFFECT

Thermal mass refers to the ability of a material to store heat. In the prototype house, materials that possess high thermal mass will absorb heat from the surrounding environment, which is particularly useful during warm periods. The stored heat is then gradually released back into the building at night, when the ambient temperature is lower.

By doing so, thermal mass counteracts temperature swings and contributes to a more stable indoor climate. Thermal mass can be used to modulate the indoor temperature and is highly effective in climates where large temperature differences occur between day and night.

In the present prototype, thermal mass is considered as a passive cooling design principle rather than as the primary experimental cooling mechanism. The main experimentally demonstrated techniques are reflective roofing, shading, and natural ventilation.

4.6 TEMPERATURE MONITORING SYSTEM

The temperature monitoring system is integrated with the passive cooling prototype to measure and observe the thermal conditions inside and outside the model house. Two DS18B20 digital temperature sensors are used for temperature measurement. One sensor is positioned inside the prototype to measure the indoor temperature, while the second sensor is positioned outside the prototype to measure the ambient outdoor temperature.

The sensors are connected to an ESP32 development board, which acts as the main controller. The ESP32 receives the temperature measurements from both sensors and processes the readings.

An OLED display is connected to the ESP32 to provide real-time visualization of the measured temperature values. The display can show the indoor temperature, outdoor temperature, temperature difference, and system status.

A DS3231 RTC module is used to provide the date and time corresponding to the temperature measurements. The SD card module is connected to the ESP32 for storing

the measured temperature data along with the corresponding time information.

The monitoring system therefore allows the thermal performance of the passive cooling prototype to be evaluated without continuously depending on a laptop for displaying the measured values.

4.7 WORKING OF ESP32-BASED MONITORING SYSTEM

The working of the monitoring system takes place in the following sequence:

Step 1: The indoor and outdoor DS18B20 sensors measure the respective temperatures.

Step 2: The temperature values are transmitted to the ESP32.

Step 3: The ESP32 processes the sensor readings.

Step 4: The RTC module provides the corresponding date and time.

Step 5: The ESP32 sends the measured values to the OLED display.

Step 6: The indoor temperature, outdoor temperature, and temperature difference can be displayed on the OLED.

Step 7: The temperature and time information are stored in the SD card for later analysis.

Step 8: The recorded data is used to prepare temperature tables and graphs.

Step 9: The indoor and outdoor temperatures are compared to evaluate passive cooling performance.

Basic calculation

$$\Delta T = T_o - T_i$$

where,

ΔT = Temperature difference

T_o = Outdoor temperature

T_i = Indoor temperature

S. No	Passive Cooling Technique	Principle Involved	Temperature Reduction (°C)	Advantages	Limitations
1	Natural Ventilation (Cross Ventilation)	Pressure difference due to wind	1 – 2	Simple, no energy consumption	Depends on wind availability
2	Stack Ventilation	Buoyancy effect of warm air	1 – 2	Effective in hot climates	Requires height difference
3	Cool Roof (Aluminium Sheet)	Solar radiation reflection	2 – 4	Reduces roof heat gain	Effect reduces at night
4	Shading Devices	Reduction of direct solar radiation	1 – 2	Low cost and effective	Needs proper orientation
5	Thermal Mass	Heat storage and time lag	2 – 3	Smooth temperature variation	Adds structural weight
6	Integrated Passive Cooling System	Combination of multiple techniques	4 – 6	Highest performance	Design complexity

Table 4.1 Passive cooling comparison techniques

S.No	Parameter	Symbol	Unit	Description
1	Outdoor Air Temperature	T_o	$^{\circ}\text{C}$	Ambient temperature outside the building
2	Indoor Air Temperature	T_i	$^{\circ}\text{C}$	Temperature inside the building
3	Heat Transfer Rate	Q	W	Rate of heat flow
4	Thermal Conductivity	k	$\text{W/m}\cdot\text{K}$	Ability of material to conduct heat
5	Surface Area	A	m^2	Area through which heat is transferred
6	Temperature Difference	ΔT	$^{\circ}\text{C}$	Difference between outdoor and indoor temperature
7	Convective Heat Transfer Coefficient	h	$\text{W/m}^2\cdot\text{K}$	Heat transfers due to air movement
8	Solar Radiation Intensity	I	W/m^2	Incoming solar energy on roof surface
9	Time	t	sec / hour	Duration of heat transfer
10	Thermal Resistance	R	$\text{m}^2\cdot\text{K/W}$	Resistance to heat flow

Table 4.2 Heat transfer parameters

4.8 PASSIVE PROTOTYPE – CALCULATIONS

1. Given Data Prototype House

House Length, $L = 650 \text{ mm} = 0.65 \text{ m}$
House Breadth, $B = 450 \text{ mm} = 0.45 \text{ m}$
Wall Height, $H = 300 \text{ mm} = 0.30 \text{ m}$

Roof Run = $225 \text{ mm} = 0.225 \text{ m}$
Roof Rise (Higher side) = 365 mm
Roof Rise (Lower side) = 300 mm
Roof Thickness, $t = 200 \text{ mm} = 0.20 \text{ m}$

Vent Opening Size = $90 \text{ mm} \times 90 \text{ mm} = 0.09 \text{ m} \times 0.09 \text{ m}$

Roof Material = Cardboard
Density of Cardboard, $\rho = 650 \text{ kg/m}^3$

2. Room Volume Calculation

Formula:

$$V = L \times B \times H$$

Calculation:

$$V = 0.65 \times 0.45 \times 0.30$$

Result:

$$V = 0.08775 \text{ m}^3$$

3. Wall Area Calculation

Front and Back Walls (2 Nos.)

$$A_1 = L \times H$$

$$A_1 = 0.65 \times 0.30 = 0.195 \text{ m}^2$$

$$\text{Total} = 2 \times 0.195 = 0.39 \text{ m}^2$$

Side Walls (2 Nos.)

$$A_2 = B \times H$$

$$A_2 = 0.45 \times 0.30 = 0.135 \text{ m}^2$$

$$\text{Total} = 2 \times 0.135 = 0.27 \text{ m}^2$$

$$\text{Total Wall Area } A = 0.39 + 0.27 \text{ } A = 0.66 \text{ m}^2$$

4. Roof Slope Angle Calculation

Rise difference:

$$\Delta h = 365 - 300 = 65 \text{ mm} = 0.065 \text{ m}$$

Formula:

$$\theta = \tan^{-1} (\text{Rise} / \text{Run})$$

Calculation:

$$\theta = \tan^{-1} (0.065 / 0.225)$$

Result:

$$\theta = 16.1^\circ$$

5. Sloped Roof Length Calculation

Formula:

$$L_r = \sqrt{(\text{Run}^2 + \text{Rise}^2)}$$

Calculation:

$$L_r = \sqrt{(0.225^2 + 0.065^2)} \quad L_r = \sqrt{0.0548}$$

Result:

$$L_r = 0.234 \text{ m (234 mm)}$$

6. Roof Surface Area Calculation

Formula:

$$A_r = \text{Roof length} \times \text{Roof width}$$

Calculation:

$$A_r = 0.234 \times 0.65$$

Result:

$$A_r = 0.152 \text{ m}^2$$

7. Roof Volume Calculation

Formula:

$$V_r = \text{Area} \times \text{Thickness}$$

Calculation:

$$V_r = 0.152 \times 0.20$$

Result:

$$V_r = 0.0304 \text{ m}^3$$

8. Roof Mass Calculation

Formula:

$$m = \rho \times V_r$$

Calculation:

$$m = 650 \times 0.0304$$

Result:

$$m = 19.76 \text{ kg}$$

9. Vent Opening Area Calculation

Formula:

$$A_v = l \times b$$

Calculation:

$$A_v = 0.09 \times 0.09$$

Result:

$$A_v = 0.0081 \text{ m}^2$$

10. Total House Volume

Formula:

$$V_t = \text{Room volume} + \text{Roof volume}$$

Calculation:

$$V_t = 0.08775 + 0.0304$$

Result:

$$V_t = 0.11815 \text{ m}^3$$

CHAPTER 5

COST ESTIMATION

S.No	Component Name	Specification	Quantity	Unit Cost (₹)	Total Cost (₹)
1	ESP 32 Development Board	Microcontroller	1	450	450
2	RTC Module	DS3231	1	45	45
3	SD Card Module	SPI Interface	1	110	110
4	OLED Display	Display output	1	150	150
5	Memory Card	8 GB Micro SD	1	350	350
6	CR2032 Coin Cell	3 V Lithium	1	15	15
7	Temperature Sensor	DS18B20	2	200	400
8	Jumper Wires	Female to Female	1 set	50	50
9	USB Data Cable	USB Type	1	50	50
10	Cardboard	Prototype Material	1	150	150
11	Glue Gun	Hot Melt Type	1	200	200
12	Aluminum Foil	Reflective Sheet	1	100	100
	Total				₹2070

Table 5.1 cost estimation

CHAPTER 6

RESULTS AND SYSTEM OUTPUTS

6.1 TEMPERATURE VARIATION RESULTS

Temperature variation results show the change in indoor temperature of the prototype house over time. Readings were taken before and after applying passive cooling techniques such as natural ventilation, cool roof application, shading, and thermal mass. The comparison of these readings clearly indicates a reduction in indoor temperature when passive cooling methods are applied.

The indoor and outdoor temperature values obtained during the experimental observation are compared to determine the thermal performance of the passive cooling prototype. The indoor temperature is measured using the DS18B20 sensor placed inside the prototype, while the outdoor temperature is measured using the sensor placed outside the prototype.

S.No	Time	Outdoor Temperature	Indoor Temperature
1	09.00 AM	29.31	27.05
2	10.00 AM	30.04	28.65
3	11.00 AM	30.54	28.12
4	12.00 PM	30.21	27.45
5	01.00 PM	29.95	27.95
6	02.00 PM	28.85	26.11
7	03.00 PM	27.98	26.44
8	04.00 PM	27.88	25.85

Table 6.1 temperature variation results

From the table, it is observed that the indoor temperature is consistently 2–3°C lower than the outdoor temperature throughout the day

6.2 SENSOR READINGS AND DATA LOGGING

The accurate measurement and recording of data are paramount to assess the performance of the proposed passive cooling system. This project utilizes DS18B20 digital temperature sensors, ESP32 development board, RTC module, OLED display, and SD card storage to monitor and record the temperature variations inside and outside the prototype house.

The DS18B20 digital temperature sensors are used in this project due to their digital output nature and ability to monitor temperature values. Firstly, one unit of DS18B20 digital temperature sensor is placed inside the prototype house to determine the temperature of the indoor environment under the proposed passive cooling technique. Secondly, the second unit of DS18B20 digital temperature sensor is utilized to measure the ambient outdoor temperature. The measured indoor and outdoor temperatures are processed by ESP32 development board.

The ESP32 development board is the central processing unit of the temperature monitoring system. It connects to DS18B20 digital temperature sensors and receives the indoor and outdoor temperature values. The ESP32 development board is also responsible for calculating the temperature difference (ΔT) between the indoor and outdoor temperatures.

An OLED display is connected to the ESP32 development board to present the monitored data, including the indoor, outdoor, and ΔT temperatures, in real-time. The main advantage of the OLED display is that it can be used to demonstrate the prototype system without relying on a laptop to view the sensor data measurements.

A Real Time Clock (RTC) is used to timestamp the temperature measurements taken by the system. This is important to store and analyze the indoor and outdoor temperatures at different times, which is useful in understanding the performance of the passive cooling system. The RTC module is powered by a CR2032 coin cell, which ensures that it maintains the correct date and time even when the system is turned off.

The temperature values are stored with their corresponding timestamps using an SD card module and 8 GB memory card. The data acquisition from the system can be made in real-time or scheduled at a specific interval with a defined data format (e.g.,

text, CSV). The collected data can be transferred to a computer for analysis, comparison, plots, and other necessary data science tasks.

Therefore, The utilization of DS18B20 sensors, ESP32, OLED display, RTC module, and SD card storage provides a compact and standalone temperature monitoring system for the passive cooling project. The indoor and outdoor temperatures collected by the system can be used as data points to evaluate the performance of the prototype house’s passive cooling techniques.

Date	Time	Sensor id	Temperature
09.01.2026	10.00 AM	S1	27.85
09.01.2026	10.00 AM	S2	29.44
09.01.2026	11.00 AM	S1	26.89
09.01.2026	11.00 AM	S2	29.84
09.01.2026	12.00 PM	S1	26.99
09.01.2026	12.00 PM	S2	30.25
09.01.2026	01.00 PM	S1	27.45
09.01.2026	01.00 PM	S2	30.55

Table 6.2 Sensors readings and data logging

6.3 RESULT TABLES AND GRAPHS

This section presents the experimental results of the passive cooling system in a clear and systematic manner using tables and graphical representations. The result tables provide numerical temperature values recorded at different time intervals, while graphs visually illustrate the comparison between outdoor and indoor temperatures. These representations help in easy interpretation of the system's performance. The temperature data recorded shows that the indoor temperature of the prototype house

remains consistently lower than the outdoor temperature throughout the observation period. This difference indicates the effectiveness of passive cooling techniques such as natural ventilation, reflective roofing, shading, and thermal mass in reducing indoor heat gain.

S.No	Time	Outdoor Temperature (°C)	Indoor Temperature (°C)	Temperature Difference ΔT (°C)
1	09:00 AM	29.31	27.05	2.26
2	10:00 AM	30.04	28.65	1.39
3	11:00 AM	30.54	28.12	2.42
4	12:00 PM	30.21	27.45	2.76
5	01:00 PM	29.95	27.95	2.00
6	02:00 PM	28.85	26.11	2.74
7	03:00 PM	27.98	26.44	1.54
8	04:00 PM	27.88	25.85	2.03

Table 6.3 Temperature difference (ΔT) analysis

The above table shows the comparison between outdoor and indoor temperatures recorded at different time intervals. The temperature difference (ΔT) indicates the effectiveness of the passive cooling techniques, where indoor temperatures remain consistently lower than outdoor conditions throughout the day.

Graphs are used to compare indoor and outdoor temperature variations with time. The plotted graph clearly shows that while the outdoor temperature reaches higher values during mid-day hours, the indoor temperature rise is controlled due to the application of passive cooling techniques. The graph provides a visual validation of the numerical results presented in the tables.

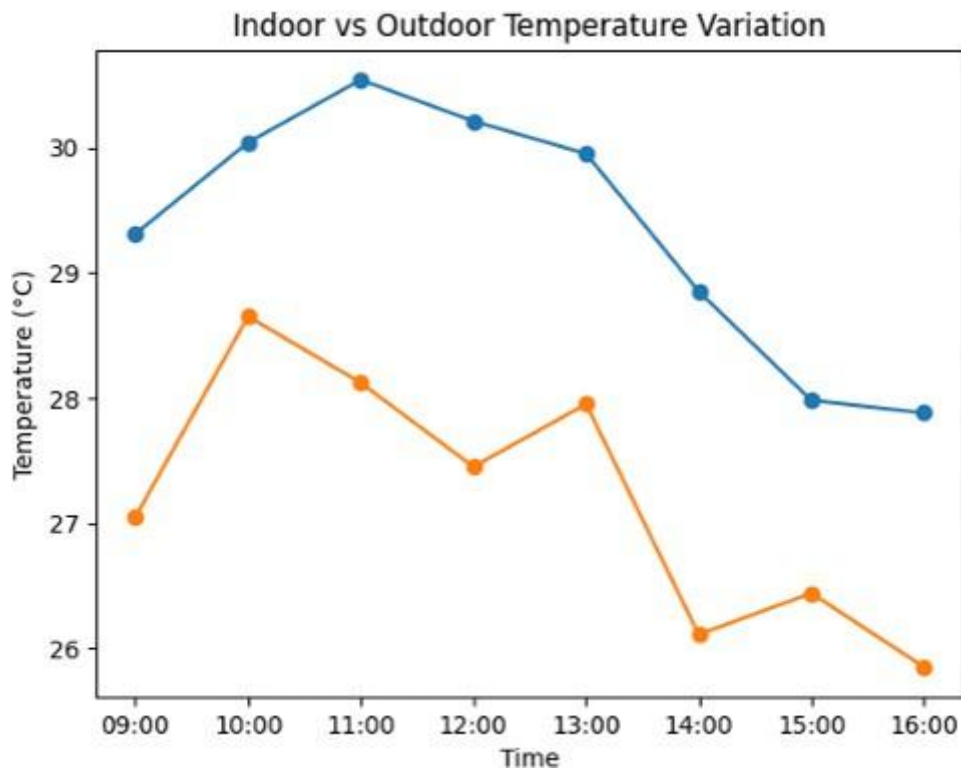


Figure 6.1 Temperature Variation Graph

The graph shows the variation of indoor and outdoor temperature with respect to time. It is observed that the indoor temperature of the prototype house remains consistently lower than the outdoor temperature throughout the day. The maximum temperature difference occurs during peak afternoon hours, indicating the effectiveness of passive cooling techniques such as natural ventilation, cool roof, and shading in reducing indoor heat gain.

CHAPTER 7

SCOPE OF FUTURE WORK

This chapter illustrates the future scope, advantages, and applications of the proposed passive cooling system. The project showcases the potential of utilizing passive cooling techniques such as natural ventilation, reflective roofing, shading, and thermal mass strategies together with an ESP32-based temperature monitoring system to minimize heat gains and enhance indoor environmental quality with limited energy use. The prototype can be modified and improved upon further to be integrated in actual practical building construction scenarios.

7.1 ADVANTAGES

The passive cooling system developed in the project has several advantages over conventional active cooling systems in use currently. One of the significant advantages of passive cooling systems is energy saving owing to the fact that the cooling is primarily achieved by utilizing passive cooling techniques such as natural ventilation, shading, reflective surfaces and thermal mass. In addition, the use of aluminium foil as a reflective roof layer will reduce the amount of solar heat gain through reflection of some of the incident solar radiation. Provision of ventilation openings will also facilitate natural air circulation, while shading will reduce the amount of direct solar radiation gain into the prototype.

The developed system is cost-effective and eco-friendly since the prototype can be constructed using easily available and inexpensive materials. In addition, a reduction in the use of electrical appliances for cooling will also reduce the associated energy use and environmental impact.

Another major advantage of the developed prototype is the use of an ESP32-based system for temperature monitoring. The system uses DS18B20 sensors to monitor both indoor and outdoor temperatures and an OLED display to provide temperature readings. In addition, the system is equipped with an RTC module for time tracking and an SD card for storage of temperature data for analysis.

The developed system is easy to operate and can be used for demonstration purposes due to its compact nature and ease of use. The system is ideal for use as a demonstrative model in academic and engineering-related fields of study to demonstrate the concept of building heat gain and loss.

7.2 APPLICATIONS

The proposed passive cooling system can be applied in residential building construction, especially in hot climate regions to reduce heat gains through the use of reflective roofing, natural ventilation, and shading. The various passive cooling techniques demonstrated in the prototype can be integrated into residential house designs to enhance indoor environmental quality.

In addition, the developed system can be applied to low-cost housing and rural homesteads since it offers an economical and energy-efficient solution to building cooling, especially in warm and hot regions. It can also be applied in small-scale buildings such as offices, class-rooms, and temporary structures.

The prototype can also be used in academic and engineering institutions as a demonstrative and educational tool to demonstrate concepts of heat gain, natural ventilation, solar heat gain, and passive building cooling techniques. The ease of operation and compact design make it ideal for use in classroom demonstrations.

The ESP32-based temperature monitoring and recording system can be utilized to monitor and record indoor and outdoor temperatures to facilitate experimental studies and future prototype development

CHAPTER 8

APPENDICES

The appendices include supplementary information relevant to the project and complement the report's main content. This section comprises details of our field visit, photographs of the prototype setup, component specifications, and the Arduino code for temperature data logging.

8.1 PSEUDOCODE FOR ESP32 BASED TEMPERATURE MONITORING AND DATA LOGGING

The pseudocode below represents the processes carried out by the temperature monitoring and data logging system that was used in conjunction with the passive cooling prototype. The system comprises an ESP32 development board, two DS18B20 temperature sensors, OLED display, RTC module, and an SD card module. One DS18B20 temperature sensor was placed inside the prototype house while the other was placed outside to monitor the respective temperatures.

Pseudocode

START

Initialize ESP32

Initialize DS18B20 temperature sensors

Initialize OLED display

Initialize RTC module

Initialize SD card module

Check the connected sensors and modules

If all required modules are available:

 Display system status on OLED

Repeat the following steps at the selected time interval:

 Read indoor temperature from DS18B20 Sensor 1

Read outdoor temperature from DS18B20 Sensor 2

Obtain current date and time from RTC module

Calculate temperature difference

$$\Delta T = \text{Outdoor Temperature} - \text{Indoor Temperature}$$

Display the following values on OLED:

Indoor Temperature

Outdoor Temperature

Temperature Difference

Date and Time

Store the following information in the SD card:

Date

Time

Indoor Temperature

Outdoor Temperature

Temperature Difference

Continue monitoring

If a sensor is disconnected:

Display sensor error/status on OLED

If SD card is unavailable:

Display SD card error/status on OLED

Continue the measurement process until the experiment is completed

Retrieve the recorded data

Analyze indoor and outdoor temperature values

Prepare temperature tables and graphs

Evaluate the performance of the passive cooling system

STOP

8.2 FIELD VISIT

As a part of the DIV2 Institute to Industry (I2I) Training Program, our project team visited Selvamani Sir's residence at Dindigul on 6 th November to gain practical insight into passive cooling systems implemented in a residential building.

The visit was indeed enlightening and inspiring since the residence was constructed based on sustainable architectural and climatic design principles. The house smartly utilizes a combination of natural resources and eco-friendly design to meet indoor thermal comfort while conserving energy.

In this visit, we were able to discover the following four green principles that were incorporated into the house:

1.Solar Energy Utilization

The house utilizes solar energy systems to supplement the demand for energy in the house hence reducing reliance on the main electricity grid.

2.Rainwater Harvesting

The house has a rainwater harvesting system that supports indoor water use hence conserving water.

3.Low Carbon Emission through Tree Plantation

The plantations around the house reduce heat gain and indoor carbon emissions.

4.Passive Cooling Techniques

The house utilizes passive cooling techniques to reduce heat gain and indoor temperatures.

This field visit enabled us to obtain first-hand experience on the application of passive cooling techniques that we have studied in this module. The visit was indeed edifying as we gained practical insight into the principles of eco-friendly building construction and sustainable housing.

Conclusively, the field experience complemented our theoretical understanding of passive cooling techniques in residential buildings.



Figure 8.1 Field Visit Images

8.3 PHOTOGRAPHS OF PROTOTYPE SETUP

In this section, photographs of the developed prototype of a passive cooling system are provided. The photos illustrate the general view of the prototype house, roof organization with aluminium foil, openings for air circulation, sensor locations, and other electronic components for temperature sensing.

These photographs of the constructed prototype can be referred to as the visual representation of the experimental procedure used in the research. Accordingly, these photos provide a greater extent of information about the layout and aid in describing the work of the system explained in previous chapters.



Figure 8.2 Prototype Setup House

CHAPTER 9

CONCLUSION

This project aimed to design, develop and evaluate a passive cooling system based on a low-cost model of an experimental house, equipped with temperature sensors and logging. As a result, it was possible to find out how to decrease the temperature in the room without using air-conditioning systems.

Thus, the use of passive cooling systems in the experiment with a model of an experimental house proved cost-effectiveness and sustainability. The use of an aluminum sheet on the roof, the proper roof design, and separate vent system, combined with the prevention of solar radiation, led to the reduction of the temperature. A natural draft of air and the choice of materials also impacted the changes. Data collected by temperature sensors and processed by Arduino/ESP system provided opportunity to record and analyze the results. Field study allowed validating the obtained results and evaluating the experience of sustainable housing building.

Therefore, the work described in this paper can be considered as a successful project, as it helped identify the effectiveness of a passive cooling system based on sustainability and low-cost. It can be especially useful in tropical climates like India to reduce the temperature in the room without using air-conditioning systems.

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